subscribers in September 2021, has said it has completed plans for coverage in 1,000 cities. Its rollout model is based on targeting customer consumption revenue via precise coverage planning. In November, Bharti Airtel conducted a 5G test in partnership with Nokia on the outskirts of Kolkata, while Vodafone Idea conducted similar trials in Gandhinagar in December. According to media reports, over the next two years, India will also see the installation of some 800,000 new mobile towers to facilitate the 5G rollout. There are also plans to connect up to 70 per cent of the towers using fibre optic cables by March 2024, up from the current 34 per cent.

A study conducted by Viavi Solutions, a US-based telecom network company, in February 2021, found that commercial 5G was available in 1,336 cities in 61 countries across the globe, a huge increase from the only 376 cities that had the technology the previous year. China topped the list of the most connected cities, followed by the US and South Korea. Other countries in the pecking order, as per the study, included the UK, Spain, Canada, Australia, Saudi Arabia, Italy and France.

The three major benefits of 5G, as touted by telecom operators and equipment providers, are its high bandwidth, its low latency or response time and the higher density of connections it allows. As far as bandwidth is concerned, while 4G handles about 200 megabytes (MB) of data per second, 5G can handle up to one gigabyte (1,000 MB) per second or higher. When it comes to response time—the time it takes to begin loading a page or a video after clicking a link—4G has a 100 millisecond delay, while 5G gets the job done in one millisecond, or essentially, in real time. 5G networks can also handle much higher traffic—allowing about 10 times as many device connections per square kilometre as existing networks.

The possibilities of such efficient and fast internet networks are enormous. In some countries, 5G will power autonomous vehicles, giving them a real-time awareness of every other vehicle on the road, as well as of traffic signals, linking them 'intelligently' across the IoT (the Internet of Things, the network of internet-connected devices that runs somewhat parallel to the Internet we use to access social media, search engines, streaming services and so on). "With a confluence of technologies (IoT, big data, AI, etc.) maturing together, 5G is set to be a game changer for telecom and is expected to yield enormous economic opportunities over the next three to five years," says K.G. Purushothaman, partner and sector head of telecom at KPMG India. According to a KPMG International survey in 2020, the arrival of enterprise 5G will impact firms in almost every sector, and has the potential to unlock $4.8 trillion (Rs 358 lakh crore) of economic potential across the globe. In India, that figure is estimated at $48.7 billion (Rs 3.6 lakh crore) by 2023, with 5G contributing some 0.35-0.50 per cent to annual GDP.

Currently, Indian telcos use spectrum in the 800, 900, 1,800, 2,100 and 2,300 MHz bands to offer 4G connectivity. These can be repurposed for 5G, while the Centre has also earmarked spectrum in the 3,300-3,600 MHz bands for the new networks. On the consumer side of matters, Bharti Airtel had in 2020 urged the DoT to issue guidelines for new 5G handsets to ensure that all new models can connect to all the new 5G spectrum bands. Aside from the higher frequencies, 5G networks also operate on different architecture—instead of relying on large towers, they organise into clusters of small cells and use a principle called 'beamforming' to boost the efficiency of transmissions.

However, the rollout of new networks has renewed concerns about the safety of cellular transmission. Critics describe even existing mobile towers as akin to running powerful, unshielded microwave ovens above residential and commercial areas, pointing out that there are no definitive studies on the effect of such radiation on human health. On the other hand, even 5G's highest frequency radiation falls into the 'non-ionising' category, unlike the x-rays and gamma rays that we know are harmful. These signals also lose power rapidly over small distances. In terms of human safety, the US CDC (Centers for Disease Control and Prevention) says, "At this time we do not have the science to link health problems to cell phone use", be it for 3G, 4G or 5G networks.

In the US, there have also been concerns around the rollout of new 5G networks because of the effect they could have on frequencies used by air traffic. The CEOs of major US passenger and cargo carriers on January 17 warned that they ran the risk of a 'catastrophic' aviation crisis when cellular carriers AT&T and Verizon deployed their new 5G services near some airports. These worries also led Air India to cancel several of its flights to that country.

What is for certain is that 5G is not simply a matter of faster internet—the scale-up in cellular capability from 4G to 5G networks could reshape how mobile devices and people connect to each other and to the internet. And while India is late to the 5G party, given the number of subscribers in the domestic market, it has a good chance of becoming a leading player in this sector. ■

WHY 5G MATTERS

$4.8 TRILLION
(Rs 358 lakh crore)
The economic potential of 5G by 2023, as per a report by KPMG

$48.7 BILLION
(Rs 3.6 lakh crore)
The economic potential of 5G in India by 2023

BENEFITS

HIGH BANDWIDTH: While 4G can handle about 200 megabytes (MB) of data per second, 5G can handle up to one gigabyte (1,000 MB) per second or more

LOW LATENCY: The response time, or the time a network takes to 'respond to a click', is 100 milliseconds on 4G networks, but just one millisecond on 5G

DENSER NETWORK: 5G networks can handle 10 times the number of devices per square kilometre that 4G can